Heart Disease Prediction using Supervised Machine Learning

1. Title

Heart Disease Prediction using Supervised Machine Learning

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2. Problem Statement

Heart disease is one of the leading causes of death worldwide. This project develops a supervised machine learning model to predict whether a patient has heart disease using clinical attributes. Multiple algorithms are compared to identify the best performing model.

3. Dataset Description

Dataset: Heart Disease Dataset (Kaggle)
Rows: 1025
Columns: 14
Target: target
Target Values:
0 = No Heart Disease
1 = Heart Disease Present

Features:
age, sex, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal and target.

4. Data Preprocessing Steps

- Loaded dataset using Pandas.
- Displayed first five rows.
- Checked dataset shape.
- Verified missing values (none found).
- Removed duplicate records.
- Selected input and output variables.
- Split dataset into 80% training and 20% testing.

• Applied StandardScaler for feature scaling.

5. Exploratory Data Analysis

EDA was performed using Pandas and Matplotlib.

Include these screenshots from your notebook:
• Dataset Preview (df.head())
• Dataset Information (df.info())
• Statistical Summary (df.describe())
• Target Distribution Graph
• Correlation Heatmap
• Feature Importance Graph

Observations:
• Dataset contains no missing values.
• Dataset is suitable for binary classification.
• Important features include cp, thalach, oldpeak, ca and thal.

6. Model Used

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. K-Nearest Neighbors (KNN)
5. Support Vector Machine (SVM)

Random Forest was selected as the best model based on overall performance.

7. Training Process

1. Load Dataset
2. Preprocess Data
3. Perform EDA
4. Split Train/Test Data
5. Scale Features
6. Train Five ML Models
7. Predict Test Data
8. Compare Results

9. Select Best Model

8. Evaluation Metrics and Results

Model	Accuracy
Logistic Regression	77.05%
Decision Tree	73.77%
Random Forest	83.61%
KNN	73.77%
SVM	78.69%

Best Model: Random Forest

Accuracy : 83.61%
Precision : 78.79%
Recall : 89.66%
F1 Score : 83.87%

Insert screenshots of:
• Model Comparison Table
• Confusion Matrix
• Classification Report

9. Conclusion

The project successfully predicts heart disease using supervised machine learning. Five algorithms were compared and Random Forest achieved the highest accuracy of 83.61%, making it the most suitable model for this dataset.

10. References

1. Kaggle Heart Disease Dataset
2. UCI Machine Learning Repository
3. Scikit-learn Documentation
4. Pandas Documentation
5. NumPy Documentation
6. Matplotlib Documentation
7. WHO - Cardiovascular Diseases